

Srivatsa M R

Email: sri1mudumba@gmail.com Ph. No: +44(0)7741787488 LinkedIn: [Srivatsa M R | LinkedIn](#)

Personal Profile

Computational scientist passionate about scientific computing, mathematical modelling, and performance engineering. Experienced in optimizing large-scale numerical codes across CPU, GPU, and distributed architectures, and in enabling research teams to achieve maximum computational efficiency through profiling, code optimization, and HPC best practices.

Professional Experience

Computational Scientist, Numerical Algorithms Group Limited (Contractor to bp HPC R&D Team)

Oxford | 2023-Present

- Profiled and optimized a large-scale parallel chemistry simulation module (C++, Python, MPI, OpenMP), achieving over 30% speedup and near-linear scaling across thousands of cores in a production HPC environment
- Developed a NUMA-aware Dask worker orchestration tool to improve parallel efficiency and reduce memory bottlenecks in distributed workloads
- Lead developer of the Asset and Well Trajectory Optimizer ([AWTO](#)) – conceived as a personal research project and scaled to a production system now used across bp for asset optimization

Scientist/Engineer SD, Flight Dynamics Group, Indian Space Research Organisation

India | 2018-2022

- Designed and implemented algorithms and numerical models for satellite orbit simulation, collision avoidance, and optimal trajectory planning, improving computational efficiency and reliability of mission planning software
- Developed in-house optimization software for geostationary station acquisition and interplanetary launch opportunity analysis, using evolutionary and multi-objective optimization methods
- Contributed to mission-critical flight dynamics operations through precise simulation, algorithmic optimization, and high-stakes software development in projects like India's lunar mission

Education

MSc Applied Computational Science and Engineering, Imperial College London

2022-2023

- Graduated with Best Overall MSc ACSE Student Award, 2022-2023
- Specialised in parallel programming, scientific computing and high-performance software design

BTech Aerospace Engineering, Indian Institute of Space Science and Technology

2014-2018

- Awarded Best Overall Performance among Undergraduate Programmes
- Focus on astrodynamics, robotics and computer vision

Publications and Presentations

Srivatsa M R, Gurpreet Singh. 2021. Optimal Impulsive Maneuver Planning for Station Acquisition of Geosynchronous Satellites. Presentation at 4th National Conference on Multidisciplinary Design, Analysis and Optimization (7th – 9th Oct 2021)

Skills

Languages & Tools: C++, Python, pybind11, OpenMP, MPI, Dask, Slurm, Linux, CUDA

Optimization & Profiling: Vtune, Valgrind, NUMA optimization